

From serial MRI scans to reproducible lesion measurements.

OncoFlow detects and segments bone-related tumors, turning complex MRI data into traceable measurements and—next—clear interval change.

●

IMPLEMENTED
single-scan foundation

○

NEXT
paired longitudinal flow

01 The clinical-research gap

Visual comparison across MRI scans is time-consuming and can make small, irregular lesions difficult to quantify consistently.

02 The OncoFlow value

A backend-first workflow that preserves imaging lineage while producing lesion-level masks and physical measurements for structured review.

Focused by design

MRI only · DICOM in · NIfTI-centered processing · paired comparison before multi-timepoint history

CORE DATA FLOW

One traceable path from image to insight

IMPLEMENTED TODAY

NEXT: PAIRED COMPARISON

Each stage produces durable artifacts or structured records for the next stage—without re-running analysis at retrieval time.



PAIRED MRI CONCEPT

Make change visible—and measurable

NEXT



OUTPUT

Δ volume

Δ diameter

match status

Conceptual fields—no efficacy or performance claim

STRUCTURED OUTPUTS

Built for machines and reviewers

- 01

Lesion mask

Reusable segmentation artifact with image lineage
- 02

Physical measurements

Volume and longest diameter per stable lesion ID
- 03

Results API

Case metadata, artifacts and machine-readable lesion records
- 04

Comparison + report

Planned interval-change bundle and review summary

NEXT

SYSTEM ARCHITECTURE

Asynchronous by default



○ PostgreSQL metadata

◇ managed image artifacts

⦿ Redis / Celery orchestration target

GUARDRAILS

Designed for responsible review

Quantitative support—not diagnosis or treatment recommendation.

01

Privacy boundary

De-identification and PHI-safe descriptors at intake and review surfaces

02

Lineage & auditability

Stable study / lesion identifiers and traceable stored artifacts

03

QC + human review

Reviewable masks and quality signals; interpretation remains with qualified users

04

Access control roadmap

Authorized result access and audit logging planned for pilot hardening